

Continual Learning, 2025–September 2026: State of the Art, Bottlenecks, and a Prior-Art Map for the CRR Equanimity Rule

TL;DR

- The field has converged, empirically, on a single dominant lever — **replay/regularisation that keeps the model's output distribution close to its prior distribution** — and on a strong *empirical* predictor of forgetting (KL divergence to the base model on the new task, "RL's Razor," Shenfeld et al., ICLR 2026), but essentially every quantitative knob (replay fraction, loss weight λ , replay-batch size, memory-bank update frequency) is **hand-set, not derived**. This is exactly the gap CRR's ratio-based, unit-free quantities target.
- Alex's "equanimity" rule (weight the mean replay gradient equal to the mean current-batch gradient, $\Omega=1$) has **no exact prior art**: the nearest neighbours are GradNorm (multi-task gradient-norm equalisation, 2018), MEGA-I (loss-ratio balancing of current vs. episodic-memory gradients, 2020), and GeRe's dynamic weight matching replay *loss magnitude* to current-task loss (2025). All balance losses or equalise norms across tasks; none states a closed-form weight-1 replay rule, and none reports the specific "matches 100%-replay at ~40% less compute" result.
- The **path-length vs. endpoint** prediction is genuinely under-tested. Current practice measures *endpoint* KL / parameter displacement (RL's Razor; Bethune et al. scaling laws) or bounds *per-step* forgetting (Loss-Adaptive Learning Rates, 2026), and there is documented "drift-then-recover" oscillation in fine-tuning — but no work uses **cumulative** Fisher-Rao/KL trajectory length as the forgetting predictor, leaving CRR's coherence-length hypothesis open.

Key Findings

1. **Replay dominates at scale, but the mixing weight is heuristic.** Across continual pretraining (Ibrahim et al., 2024), LLM fine-tuning (Marek et al., 2026; Kotha & Liang, 2026), and classic class-IL (DER++, Buzzega et al., 2020), replay is the workhorse. Marek et al. use a *fixed* replay weight $\lambda=10$ on a replay minibatch $4\times$ smaller than the downstream batch; DER/ER use fixed per-benchmark α/β on a separately-drawn buffer minibatch; continual pretraining uses hand-set 5–50% replay fractions. None of these weights is derived.
2. **The forgetting "law" is empirical, not derived.** RL's Razor (Shenfeld, Pari & Agrawal, ICLR 2026, arXiv 2509.04259) shows forgetting collapses onto a single curve when plotted against forward KL(basel|fine-tuned) on the new task. The fit is far tighter in a controlled toy setting than in LLMs: "A quadratic fit achieves $R^2 = 0.96$ in this setting [ParityMNIST]... The same correlation appears in our LLM experiments, with a quadratic fit achieving $R^2 = 0.71$." On-policy RL is implicitly biased toward KL-minimal solutions. The *law itself* (KL predicts forgetting) is fitted; the "why RL stays KL-small" part is derived.

3. **Loss of plasticity has no consensus cause.** Competing 2024–2026 accounts: unit dormancy / lack of injected diversity (Dohare et al., Nature 2024, continual backpropagation); Hessian/eNTK **spectral collapse** (He et al., ICML 2026); dynamical-systems trapping near stable manifolds (Joudaki et al., ICLR 2026); and — newly — **abundant replay memory itself suppressing plasticity** (Cho et al., "Forget Forgetting," ICLR 2026). Each proposes a different fix, which is itself evidence the mechanism is unsettled.
4. **Multi-timescale memory is back, with hand-set frequencies.** Google's Nested Learning / HOPE (Behrouz et al., arXiv 2512.24695) frames a model as nested optimisation problems and a "Continuum Memory System" of memory banks each updating at a *different, hand-specified frequency* — a direct architectural analogue of CRR's occasions/resets, but with the timescales chosen by hand rather than derived from the data's own resolvable steps.
5. **Mechanistic localisation of forgetting is converging on middle/attention layers but is contested.** 2026 mechanistic work reports representational drift in intermediate layers, gradient interference in attention weights, and MoE routing-gate drift; but the flagship arXiv 2601.18699 has hallmarks of an unreliable source (see Caveats).

Details

PART 1 — State of the Art (with mechanism, and whether the explanation is derived or empirical)

Replay methods.

- *Experience Replay (ER)* and *Dark Experience Replay (DER/DER++)* (Buzzega et al., NeurIPS 2020, arXiv 2004.07211). Mechanism (authors): interleaving buffered exemplars — and, in DER, matching stored logits sampled along the optimisation trajectory (knowledge distillation) — keeps the function from drifting on past data; DER++ adds a ground-truth-label term to survive sudden distribution shift. (Liner) DER uses reservoir sampling and needs no task boundaries. (NeurIPS) *Empirical*. The buffer minibatch is drawn separately and its loss added with fixed coefficients α (logit-matching) and β (labels), (NeurIPS) typically $\alpha=1$, $\beta=0.5$, buffer minibatch ≈ 32 .
- *CLS-ER* (dual-memory experience replay, Arani et al., ICLR 2022): maintains fast- and slow-updating semantic memories (EMA of weights) alongside the episodic buffer; mechanism (authors) is the complementary-learning-systems analogy — a plastic fast learner and a stable slow learner, consolidated by aligning decisions on buffered samples. *Empirical/analytical*.
- *SuRe (Surprise-driven Prioritised Replay)* (Hazard et al., 2025, arXiv 2511.22367): prioritises replay of high-surprise (high prediction-error) prompts. Mechanism: surprise proxies for samples most at risk of being overwritten; concentrating replay there is sample-efficient. *Empirical*.
- *Self-generated / synthetic replay*: the model samples its own training distribution instead of storing exemplars. In LLMs this "nearly eliminates forgetting" (Marek et al.,

2026, arXiv 2605.26097); (arXiv) Shin et al. (2017) deep generative replay is the origin. Mechanism (authors): self-samples approximate the pretraining distribution well enough that regularising on them holds the prior distribution fixed. *Empirical, with a partial derivation* — Marek et al. show NTP-on-self-samples is a Monte-Carlo estimator of forward KL to the base model. (arxiv)

- *On-policy replay for continual SFT* (arXiv 2605.29495, 2026): replaces stale gold answers attached to historical prompts with reward-filtered on-policy samples. Mechanism: on-policy targets stay near the current policy, reducing distributional drift (a corollary of RL's Razor). *Empirical, motivated by theory*.
- *GeRe (General-sample Replay)* (Zhang et al., 2025, arXiv 2508.04676): a small, *fixed* pre-collected set of generic pretraining texts is replayed to retain general capability, (arXiv) with a threshold-based-margin (TM) activation-consistency loss. Mechanism (authors): anchoring neural activation states on general samples preserves broad capability and even improves downstream performance. (arXiv) *Empirical*. Notably, GeRe uses **Dynamic Weight Balancing**: $\omega^{\text{TM}} = \text{detach}(L^{\text{CE}} / L^{\text{TM}})$ so the replay loss magnitude is scaled to match the current-task CE loss (arxiv) — the closest published analogue to an "equanimity" idea, but on losses, not gradients.

Regularisation and distillation.

- *EWC family* (Kirkpatrick et al., PNAS 2017) and *Synaptic Intelligence* (Zenke et al., 2017): quadratic penalty on parameter change weighted by (Fisher-based) importance. Mechanism (authors): protect parameters important to prior tasks. *Derived* (Laplace/Fisher approximation) but the approximation is known to be loose; RL's Razor reframes EWC as an approximate KL-minimisation.
- *Learning without Forgetting* (Li & Hoiem, 2017): distil the old model's outputs on new-task inputs. Mechanism: functional (output-space) anchoring. *Empirical*.
- *KL regularisation on replay data* (Marek et al., 2026, arXiv 2605.26097): add $\lambda \cdot \text{KL}(\text{base} \parallel \text{model})$ on replay/self-generated data; they find the KL objective beats plain NTP for retention, (arxiv) and that $\lambda=10$ (verbatim, Fig. 4/6 captions: "a strong regularization coefficient ($\lambda=10$) to minimize forgetting") on a **4×-smaller replay minibatch** (Appendix B.2: "a batch size 4-times smaller than the downstream batch (to reduce the computational overhead from regularization)") works well. Capacity lower-bounds forgetting: a near-saturated (overtrained) model *cannot* learn without overwriting. (arXiv) *Empirical, with the KL-estimator derivation above*. λ and the batch ratio are hand-set.
- *Self-Distillation Fine-Tuning (SDFT)* / "Self-Distillation Enables Continual Learning" (Shenfeld, Damani, Hübötter, Agrawal, 2026, arXiv 2601.19897; ICML 2026): use in-context conditioning to turn a demonstration into a per-instance reward, enabling *on-policy* learning from demonstrations. Mechanism (authors): on-policy distribution alignment keeps updates in a trust region around the base policy → less forgetting than off-policy SFT. *Derived from an inverse-RL argument*.

- *Self-Distillation Policy Optimization (SDPO)* / "Reinforcement Learning via Self-Distillation" (Hübotter et al., 2026, arXiv 2601.20802): the verifiable-reward analogue of SDFT; achieves a better performance-forgetting trade-off than GRPO. *Empirical + theory.*
- *RL's Razor* (Shenfeld & Agrawal, ICLR 2026, arXiv 2509.04259): the empirical forgetting law KL(baselfine-tuned) on the new task predicts forgetting; on-policy RL is implicitly biased toward KL-minimal solutions among those solving the task. (arXiv) They ablate weight change, representation drift, rank/sparsity, reverse KL, total variation, L2 — none explains forgetting as well as forward KL. *Law is empirical ($R^2=0.96$ toy / 0.71 LLM); the "why RL is KL-minimal" part is derived.*
- *Critique — "Rethinking On-Policy Self-Distillation for Thinking Models"* (arXiv 2607.05184): questions whether on-policy self-distillation retains its advantage on long-chain-of-thought "thinking" models. [Flagged: I could not independently verify this arXiv ID within budget; treat as claimed.]

EMA / dual-memory / slow-fast consolidation. Mean-teacher/EMA weight averaging and CLS-ER (above) instantiate a stable slow copy and a plastic fast copy; "Forget Forgetting" (Cho et al., ICLR 2026) uses weight averaging for stability + rank-based resets for plasticity. Mechanism: temporal averaging suppresses high-frequency drift. *Empirical/analogue.*

Multi-timescale architectures. *Nested Learning / HOPE / Titans continuum memory* (Behrouz et al., Google, arXiv 2512.24695): memory as a spectrum of banks, each updating at a hand-set frequency; (Google Research) HOPE adds self-modifying ("unbounded levels of in-context learning") updates on top of Titans' surprise-based long-term memory. (Medium) Mechanism (authors): separating timescales lets fast banks absorb novelty while slow banks consolidate, reducing interference. *Empirical (proof-of-concept), timescales hand-set.*

Continual pretraining of LLMs.

- *Ibrahim/Gupta et al.* (TMLR 2024, arXiv 2403.08763): verbatim, "a simple and scalable combination of learning rate (LR) re-warming, LR re-decaying, and replay of previous data is sufficient to match the performance of fully re-training from scratch on all available data... matching the re-training baseline using only a fraction of the compute" — validated at 405M and 10B across weak ($En \rightarrow En$) and strong ($En \rightarrow De$) shifts. Replay fraction is hand-set (small % suffices for weak shift; more for strong shift). *Empirical.*
- *Kotha & Liang, "Replaying pre-training data improves fine-tuning"* (2026, arXiv 2603.04964): replaying generic data during fine-tuning not only prevents forgetting but *improves* the target task — up to $1.87\times$ (fine-tuning) / $2.06\times$ (mid-training) data efficiency; helps more when target data is scarce in pretraining. *Empirical.*
- *"The Finetuner's Fallacy"* (Baek et al., 2026, arXiv 2603.16177): specialised pretraining (repeating the domain set from pretraining) beats fine-tuning-only; a 1B specialised-

pretrained model can beat a 3B standard model on narrow domains, with less forgetting. Replay rates {0,10,20%} tuned by hand. *Empirical*.

- "*Sharpness-Aware Pretraining Mitigates Catastrophic Forgetting*" (Watts et al., 2026, arXiv 2605.02105): SAM / large LR / short anneal bias toward flat minima; a short SAM mid-training phase on OLMo-2-1B cuts forgetting by 31% (MetaMath) and 40% (4-bit quant); up to 80% less forgetting across 20M–150M models. Mechanism: flatter base minima leave more capability intact under later updates. *Empirical, geometry-motivated*.
- *Scaling laws for forgetting* (Bethune et al., 2025, arXiv 2502.06042): forgetting under pretraining-data injection follows a scaling law in injection fraction and model size; uses endpoint cross-entropy-to-base as the forgetting metric. *Derived scaling fit (empirical form)*.

Loss of plasticity.

- *Dohare et al.* (Nature 632:768–774, 2024): standard deep nets lose plasticity in continual settings — verbatim, "In ImageNet, binary classification performance dropped from 89% accuracy on an early task down to 77%, about the level of a linear network, on the 2000th task." Continual backpropagation (reinitialise a small fraction of least-used units) maintains plasticity. Mechanism (authors): sustained learning needs a random, non-gradient diversity injection. *Empirical*.
- "*Spectral Collapse Drives Loss of Plasticity*" (He et al., ICML 2026, arXiv 2509.22335): plasticity loss is preceded by Hessian spectral collapse at new-task init; derives ϵ -rank conditions and shows the loss-weighted Gram matrix is spectrally equivalent to the Gauss–Newton approximation; fix = maintain effective feature rank + L2. *Derived*.
- *Activation-function design for plasticity* — deep Fourier features and learnable/wavelet activations (2026, e.g. arXiv 2509.22562 and related): structured/periodic features resist rank collapse. *Empirical + partial theory*. (Exact IDs for the "deep Fourier features" and Aug-2026 wavelet papers not fully verified within budget.)
- "*Forget Forgetting*" (Cho et al., ICLR 2026, arXiv 2502.07274): in the abundant-memory regime the bottleneck flips from stability to *plasticity* — abundant exemplar memory suppresses new-task learning because the expected update direction collapses onto the span of past gradients as replay weight $\lambda \rightarrow 0$. Fix = Weight-Space Consolidation (rank-based resets + weight averaging). *Empirical with a short gradient-geometry derivation*.

Theory in scaling limits.

- "*The Importance of Being Lazy*" (Graldi et al., ICML 2025, arXiv 2506.16884): using DMFT, width helps only insofar as it induces *laziness* (less feature learning); a task-similarity-modulated transition separates a lazy low-forgetting regime from a rich high-forgetting regime; optimal performance at a critical level of feature learning. *Derived*.

- "Convergence of continual learning in homogeneous deep networks" (2026) and related (Jung et al., ICLR 2025 on GD implicit bias in continual linear classification; Karpel et al., 2026, optimal L2 in high-dim continual linear regression, arXiv 2601.13844): characterise convergence/implicit bias in idealised settings. *Derived*.

Gradient-balancing / gradient-projection prior art (relevant to equanimity).

- *GEM* / *A-GEM* (Lopez-Paz & Ranzato, 2017; Chaudhry et al., 2019): project the current gradient so it does not increase loss on buffered tasks. *Derived (constrained QP)*.
- *MEGA-I/II* (Guo et al., NeurIPS 2020, arXiv 1909.11763): unify GEM/A-GEM as balancing two gradients; (ML Anthology) MEGA-I sets mixing weights from the *loss ratio* (if current loss is small, shift emphasis to memory); (arxiv) MEGA-II rotates the current gradient toward the reference. GEM/A-GEM are degenerate cases that always weight the current task at 1. (ML Anthology) *Derived optimisation view*.
- *GradNorm* (Chen et al., ICML 2018, arXiv 1711.02257): equalises per-task gradient norms to the average gradient norm (Proceedings of Machi...) — verbatim, "The common scale for gradients is most naturally the average gradient norm, $G_W(t)$," and GradNorm "directly tun[es] the gradients to equalize task training rates," controlled by a single asymmetry hyperparameter α ($\alpha=0 \rightarrow$ exactly equal norms). Multi-task, not replay. *Derived heuristic*.
- *Mammoth's additive-loss ER*: separate buffer minibatch, fixed additive weight — see DER above.

Benchmarks and evaluation. Mammoth (aimagelab) is the de-facto CL framework; (GitHub) standard splits are class-IL / task-IL / domain-IL, plus task-free and recurring streams. "Continual evaluation / the stability gap" (De Lange, van de Ven & Tuytelaars, ICLR 2023, arXiv 2205.13452) shows "severe forgetting of past tasks occurs already during the first few training iterations on a new task," but this forgetting "is temporary and followed by a phase of performance recovery... we refer to [this] as the stability gap" — an endpoint-vs-trajectory distinction directly relevant to CRR. "Continual learning should move beyond incremental classification" (2025) argues the field's benchmarks are too narrow. There is a NeurIPS 2026 workshop on test-time continual-learning agents. *[Flagged: exact citations for the position paper and the NeurIPS 2026 workshop could not be fully verified within budget.]*

Agent-memory infrastructure. Continuum-Memory-System deployments for LLM agents (e.g., arXiv 2605.29055, 2601.13186) implement fast/medium/long memory layers with explicit staleness/decay; these are engineering claims, largely un-benchmarked against CL metrics.

PART 2 — Bottlenecks the field names (consolidated)

1. **The forgetting law is empirical/underived.** RL's Razor fits $KL \rightarrow$ forgetting ($R^2=0.96$ toy, 0.71 LLM) but does not derive the functional form; it explicitly frames KL as a "reliable predictor," not a first-principles law.

2. **Hand-set replay weight/loss coefficient.** Marek et al. use $\lambda=10$ on a 4 $\times$ -smaller replay batch — both chosen, not derived. GeRe uses a detached loss-ratio. DER uses per-benchmark α/β .
3. **Hand-set replay fractions in continual pretraining.** Ibrahim et al. use small % for weak shift, more for strong shift; Finetuner's Fallacy tunes {0,10,20%}; surveys note "5% vs 25% by shift strength." No derivation of the correct fraction.
4. **Hand-set update frequencies in continuum-memory systems.** Nested Learning/HOPE's memory-bank frequencies are design choices.
5. **No consensus cause of loss of plasticity** (dormancy vs spectral collapse vs dynamical trapping vs abundant-memory suppression).
6. **Capacity measured only by proxy.** Marek et al. show capacity lower-bounds forgetting but proxy it via tokens-per-parameter / pretraining saturation, not a direct measure. (arxiv)
7. **Single-task/short-sequence results may not extend to long multi-task streams** (widely acknowledged; e.g., mechanistic and post-training-forgetting papers test few tasks).
8. **Abundant memory reduces plasticity** (Cho et al.) — a newly named trade-off.
9. **Cost of on-policy sampling.** SDPO/on-policy distillation require generating rollouts (SDPO $\sim 2\times$ generations vs SFT for the self-teacher baseline); on-policy replay pays a sampling tax.
10. **Task-boundary detection in task-free streams** remains unsolved (DER avoids boundaries but general task-free CL is open).
11. **Where forgetting localises is contested** (middle layers / attention / MoE routing vs function vectors).
12. **Compute cost of replay at frontier scale.** Continual pretraining saves vs full retraining (Ibrahim et al.); Hickok (2025, arXiv 2505.12512) reports 55–65% fewer replay samples for equal performance via reduced replay ratio + consolidation. Exact frontier-scale FLOP figures for replay are rarely reported.

PART 3 — Prior-art check for the equanimity rule (honest)

- (i) **Set replay weight so replay gradient norm = current-task gradient norm / balance mean gradients of old and new data.** *No exact match found.* Near-misses: **GradNorm** (equalises task gradient norms to the average, multi-task, learns weights via auxiliary loss — not replay, not weight-1, not closed-form); **MEGA-I** (balances current vs. episodic-memory gradients by *loss ratio*, adaptive, not norm-equality); **GeRe** (dynamic weight matching replay *loss magnitude* to current CE loss via a detached ratio). All balance losses or equalise norms across tasks; none states the mean-replay-gradient = mean-current-gradient (weight-1) rule.
- (ii) **Replay minibatch size vs replay loss weight trade-off.** Partially present but not as a principled trade-off: Marek et al. use a 4 $\times$ -smaller replay batch *and* $\lambda=10$ (they up-weight the smaller batch), motivated only as "to reduce the computational overhead

from regularization." DER draws a fixed-size buffer minibatch (~ 32) with fixed α/β . No paper isolates "small batch, unit weight" as an optimum.

- **(iii) Additive-loss ER (separate means, weight 1) tolerates a much smaller replay minibatch.** *Not found as an explicit claim.* Mammoth/DER use additive separate-minibatch losses but with tuned weights, not weight 1, and do not report tolerance to shrinking the replay minibatch at unit weight.
- **(iv) Compute reduction from small-weighted replay.** *The specific "~40% less compute, matches 100% replay" is not in the literature.* Closest: Hickok (2025) — 55–65% fewer replay samples for equal performance (via reduced replay ratio + a consolidation phase, not a gradient-balancing weight). Marek et al. motivate the 4 $\times$ -smaller batch as compute-saving but give no % figure.

Verdict: The equanimity rule as Alex states it — a *closed-form, unit-free, weight-1* balancing of *mean gradients* of replay and current batches, plus the small-replay-batch compute result — appears **genuinely novel** relative to GradNorm/MEGA/GeRe/DER. The strongest overlapping ideas are conceptual (balance old vs new; equalise magnitudes), not identical.

PART 4 — Path-length vs endpoint forgetting

- **Endpoint predictors dominate.** RL's Razor uses *endpoint* forward KL on the new task; (Hugging Face) Bethune et al. use endpoint cross-entropy-to-base; many use $\| \theta_T - \theta_0 \|$. These are displacement/endpoint quantities.
- **Per-step (not cumulative-path) bounds exist.** "Fine-Tuning Without Forgetting via Loss-Adaptive Learning Rates" (2026, arXiv 2605.20005) bounds *per-step* forgetting and notes bounding it "at each step directly controls cumulative forgetting" — the closest to a path-integral view, but it bounds a sum of per-step losses, not Fisher-Rao arc length.
- **Oscillation / "wasted travel" is documented but not costed as path length.** The stability-gap literature (De Lange et al., 2023) and "Overcoming Forgetting with Evolution Strategies" (arXiv 2605.30148) show prior-task accuracy drops then recovers, and that random-walk weight drift (larger $\| \theta_t - \theta_0 \|$) tracks forgetting — evidence that trajectories oscillate and that *travel* correlates with forgetting, but nobody integrates arc length.
- **No work uses cumulative KL / cumulative Fisher distance / trajectory length as the forgetting predictor.** RL's Razor even shows per-step KL change anti-correlates with gradient similarity (bigger steps $\rightarrow$ more forgetting), which is *suggestive* of a path-length effect, but the paper's headline predictor remains endpoint KL.

Verdict: CRR's path-length (arc actually travelled) vs endpoint-displacement distinction is an **open, untested prediction**. The cheapest decisive test: fine-tune two ways to the *same endpoint KL* but different cumulative trajectory length (e.g., high-LR-oscillatory vs low-LR-direct), and check whether forgetting tracks cumulative Fisher-Rao length or endpoint KL.

Bottleneck ↔ CRR-derived-candidate table

Bottleneck	Who names it	Quantity currently hand-set	What a CRR-derived alternative must predict	Cheapest test
Forgetting law is fitted	Shenfeld et al. (RL's Razor, ICLR 2026)	Functional form of KL→forgetting ($R^2=0.96$ toy, 0.71 LLM)	Forgetting as a derived function of coherence C (Fisher-Rao arc length since reset), unit $\Omega=1$	Re-plot RL's Razor data with cumulative Fisher length as x-axis; test if R^2 improves
Replay loss weight	Marek et al. 2026 (2605.26097); DER (2020); GeRe (2508.04676)	$\lambda=10$; DER α/β ; GeRe detached loss-ratio	Weight = 1 when replay & current mean gradients are equalised (equanimity), derived not tuned	Split-MNIST/KMNIST: sweep λ , confirm optimum at gradient-norm parity; check LR/batch invariance
Replay fraction in CPT	Ibrahim et al. 2024; Baek et al. 2026	5-50% by shift strength	Fraction set so replay arc length = new-task arc length per step	Continual pretrain at matched gradient-parity fraction vs tuned fraction; compare retention/compute
Memory-bank update frequency	Behrouz et al. (Nested Learning, 2512.24695)	Hand-set per-bank frequencies	Reset/regeneration cadence derived from the system's own resolvable steps ($\Omega=1$)	Ablate: derived cadence vs hand-set on continual LM benchmark
Cause of plasticity loss	Dohare (Nature 2024); He et al. (ICML 2026); Cho et al. (ICLR 2026)	Reset fraction / rank threshold / λ	Rupture (reset) triggered when coherence exceeds a derived bound	Compare CRR-timed resets vs continual-backprop random resets on Continual ImageNet
Abundant memory suppresses plasticity	Cho et al. (ICLR 2026)	Replay weight λ (collapse as $\lambda \rightarrow 0$)	Bounded regeneration strength κ that keeps plasticity while retaining	Vary κ ; check plasticity (new-task acc) vs stability frontier

Bottleneck	Who names it	Quantity currently hand-set	What a CRR-derived alternative must predict	Cheapest test
Endpoint vs path forgetting	RL's Razor; De Lange et al. (2023)	Endpoint KL / $\ \theta_T - \theta_0\ $	Forgetting $\propto$ cumulative Fisher-Rao arc length, not endpoint displacement	Same-endpoint-KL, different-path fine-tunes; regress forgetting on arc length
Ratios vs absolute rates across systems	(CRR novel; not named as such)	Absolute forgetting rates	Retention <i>ratios</i> in the system's own metric are invariant across systems	Measure ratio-based retention across MNIST/KMNIST/Fashion-MNIST; test invariance

Recommendations

1. **Publish the equanimity rule now as a distinct contribution**, positioning it explicitly against GradNorm (norm-equalisation, multi-task), MEGA-I (loss-ratio, episodic memory), and GeRe (loss-magnitude matching). The novelty is the *gradient-mean* parity at *unit weight*, its LR/batch invariance, and the small-replay-batch compute result — none of which the three have. Report the Fashion-MNIST counterexample prominently; reviewers will trust the claim more.
2. **Run the path-length test next** — it is the cheapest high-value experiment and the one prediction with *no* existing competitor. Fine-tune to matched endpoint KL along a short-direct vs long-oscillatory path and regress forgetting on cumulative Fisher-Rao length. If forgetting tracks arc length better than endpoint KL, that is a direct, publishable refutation of the RL's-Razor endpoint predictor.
3. **Re-analyse existing public datasets** (RL's Razor release; Marek et al. code at github.com/martin-marek/forgetting; Mammoth logs) with CRR quantities before running new large models — you can test the ratio-invariance and coherence-length claims on data already released.
4. **Frame CRR against the "hand-set knob" bottleneck**, which recurs in every subfield (λ , replay fraction, memory frequency, reset fraction). A framework that *derives* any one of these from a unit-free coherence quantity is immediately useful; lead with replay weight since you already have the equanimity result.

Benchmarks that would change these recommendations: if a 2026 paper appears that (a) derives the replay weight from gradient parity, or (b) uses cumulative Fisher/KL length as a forgetting predictor, the novelty claims weaken — monitor arXiv cs.LG for both. If the equanimity optimum fails to hold on a third dataset family beyond Fashion-MNIST, downgrade recommendation 1 to "robust heuristic" rather than "derived optimum."

Caveats

- **Future-dated arXiv IDs.** Many cited papers carry 2026 arXiv IDs (26xx.xxxxx) consistent with the stated current date of September 2026. I verified the abstract/venue for the central ones (RL's Razor 2509.04259; Marek et al. 2605.26097; Nested Learning 2512.24695; Kotha & Liang 2603.04964; Finetuner's Fallacy 2603.16177; Sharpness-aware 2605.02105; Spectral Collapse 2509.22335; Forget Forgetting 2502.07274; GeRe 2508.04676; SuRe 2511.22367; Galdi et al. 2506.16884; SDFT 2601.19897; SDPO 2601.20802; Ibrahim et al. 2403.08763; MEGA 1909.11763; GradNorm 1711.02257; DER 2004.07211; stability gap 2205.13452).
- **arXiv 2601.18699 ("Mechanistic Analysis of Catastrophic Forgetting")** shows internal inconsistency across its own versions (the abstract lists twenty models including implausible names like "Claude Fable 5" and "GPT-5.5 High," while the HTML body describes six models 109B-1.5T with a Pearson $r=0.87$ task-similarity correlation). Treat its specific numbers as **unverified/possibly synthetic**; the qualitative claim (middle-layer/attention/MoE-routing localisation) is corroborated by other 2026 work but the paper itself is a low-confidence source.
- **Could not fully verify within budget:** the exact citation for "Continual learning should move beyond incremental classification" (2025), the NeurIPS 2026 test-time-CL-agents workshop, the "Rethinking On-Policy Self-Distillation" critique (arXiv 2607.05184), and the precise "Plastic learning with deep Fourier features" / Aug-2026 learnable-wavelet-activation IDs. These are reported as *claimed*.
- **The "~40% less compute" and "matches 100%-replay" equanimity results are Alex's own findings;** I found no external corroboration, and the nearest published compute-reduction number is Hickok (2025)'s 55–65% fewer replay samples via a different mechanism. This strengthens the novelty claim but means the result is currently unreplicated externally.
- **Established vs claimed vs contested:** *Established* — replay mitigates forgetting; continual pretraining saves compute vs retraining; plasticity loss is real (Nature 2024). *Claimed* — RL's Razor's KL law (strong evidence, single-curve fits, but empirical); self-generated replay "nearly eliminates" forgetting. *Contested* — the mechanism/localisation of forgetting and the cause of plasticity loss.

References (arXiv IDs and dates)

- Buzzega et al., *Dark Experience for General Continual Learning (DER/DER++)*, NeurIPS 2020, arXiv 2004.07211.
- Lopez-Paz & Ranzato, *GEM*, 2017; Chaudhry et al., *A-GEM*, 2019.
- Guo et al., *Improved Schemes for Episodic Memory-based Lifelong Learning (MEGA-I/II)*, NeurIPS 2020, arXiv 1909.11763.
- Chen et al., *GradNorm*, ICML 2018, arXiv 1711.02257.
- Kirkpatrick et al., *EWG*, PNAS 2017; Li & Hoiem, *Learning without Forgetting*, 2017.
- Arani et al., *CLS-ER*, ICLR 2022.

- Hazard et al., *SuRe*, 2025, arXiv 2511.22367.
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